

Josea Mari T. Gacutno

AI / Software Engineer

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ABOUT ME

Computer Engineering graduate of the University of the Philippines Diliman with hands-on experience building and integrating AI-powered software systems — from retrieval-augmented Q&A pipelines to full-stack automated platforms. Built a Developer Documentation RAG Q&A service using Python, FastAPI, LangChain, ChromaDB, and Ollama, including an LLM-as-judge evaluation pipeline. Co-developer of PaTsek, a multimodal automated grading platform recognized as a Top 18 Semi-Finalist for the UPEEEAI Medal for Best Undergraduate Projects 2026, where I owned end-to-end system integration and REST API development. Automated multi-platform data collection systems using Python, Playwright, and Selenium, and tested software through a Quality Assurance internship at DOST. Background also includes embedded systems work with PIC32CM microcontrollers and Xilinx FPGA platforms. Seeking a software engineering role building, testing, and integrating AI-driven production systems.

EDUCATION

Bachelor of Science in Computer Engineering

2022 – 2026

University of the Philippines Diliman (UPD) — Quezon City, Metro Manila

CERTIFICATIONS

- Zapier Academy — Building AI Agents, Governing Zapier MCP, Monitoring and Operations, Security and Governance, Account Setup, Building Intermediate Zaps (July 2026)
- Make Academy — Make Foundation, Make Basics (July 2026)

SKILLS

- **Technical Skills:** Software Development, REST API Development, Systems Integration, AI/LLM Pipeline Development, Workflow Automation, Quality Assurance & Testing, Data Validation, Embedded Systems Design
- **Soft Skills:** Analytical Problem-Solving, Cross-functional Collaboration, Technical Documentation, Project Coordination, Attention to Detail, Time Management

TOOLS & TECHNOLOGIES

- **Programming Languages:** Python, C, MATLAB, Verilog, Rust, SQL
- **AI & LLM Development:** LangChain, ChromaDB, Ollama, FastAPI, LLM-as-Judge Evaluation Pipelines
- **Software Testing & QA:** Regression Testing, Functionality Testing, Defect Identification & Documentation, Data Validation
- **Workflow Automation:** Zapier (Certified), Make (currently building hands-on proficiency), Playwright, Selenium
- **Cloud & Version Control:** Google Cloud, Git, REST APIs, Webhooks
- **Development Environments & Platforms:** MPLab, Vivado, Vitis, Visual Studio Code, Jupyter, GitHub
- **Embedded Systems & Hardware:** PIC32CM Microcontroller, Xilinx Arty A7-35T FPGA, MicroBlaze V Soft Processor, AXI4 Interconnect, I2C, PS/2, GPS Integration

EXPERIENCE

AI & Software Development Experience

Developer Documentation RAG Q&A Service

Jun 2026 – Present

Personal Project — Remote

- Built a backend pipeline for natural-language Q&A over developer documentation using Python, FastAPI, LangChain, ChromaDB, and Ollama, including an automated LLM-as-judge evaluation pipeline and a FastAPI microservice with usage and latency monitoring.

Software & AI Projects Contributor

Nov 2025 – May 2026

Digital Signal Processing (DSP) Lab, UPD EEEI — Quezon City, Metro Manila

- Owned end-to-end application development and system integration of PaTsek, a multimodal automated grading platform — connecting OMR, handwriting recognition, and component detection modules into a single cohesive application, coordinating across faculty researchers and lab members for 6+ months.

- Built and validated REST API endpoints connecting the platform to a cloud-based backend, debugging integration issues across multiple processing modules to ensure reliable end-to-end data flow.
- Contributed to model training and systematically tested and validated outputs against ground-truth data, identifying gaps and proposing optimizations — recognized as a Top 18 Semi-Finalist for the UPEEEAAI Medal for Best Undergraduate Projects 2026.
- Authored technical documentation covering system architecture, data pipelines, and operational workflows.

Social Media Analytics Automation System

Dec 2025 – Jun 2026

UP CAPES — Quezon City, Metro Manila

- Built a multi-platform automation system using Python, Playwright, and Selenium, reducing manual data collection effort by 70% through end-to-end workflow automation.

Can Satellite Project

Feb 2025 – May 2025

UPD EEEl — Quezon City, Metro Manila

- Developed a PIC32CM microcontroller-based satellite prototype with GPS parsing and wireless telemetry, including a custom web interface for real-time data visualization.

FPGA-Based I2C LCD & PS/2 Keyboard Interface

Nov 2025 – Dec 2025

UPD EEEl — Quezon City, Metro Manila

- Engineered a hardware-software embedded system integrating a custom Verilog PS/2 keyboard controller with C-based I2C LCD firmware, debugging integration issues to achieve reliable end-to-end operation.

Internship Experience

Quality Assurance (QA) Intern

Jun 2025 – Aug 2025

Department of Science and Technology (DOST) – Project LODI — Remote

- Conducted regression and functionality testing for software feature validation, systematically identifying, documenting, and tracking system defects to ensure product reliability.
- Collaborated with cross-functional development and QA teams to troubleshoot issues, improve data validation processes, and support quality assurance workflows.

University & Leadership Experience

IT Automations Associate

Aug 2025 – Present

UP CAPES — Quezon City, Metro Manila

- Automated data collection for social media analytics using Python, reducing manual effort in tracking member engagement metrics across platforms.

Vice President for Membership

Aug 2024 – Aug 2025

UP Circuit — Quezon City, Metro Manila

- Managed project activities for a full year including estimation, planning, and stakeholder management, leading recruitment strategy that resulted in 50+ new members — the highest post-pandemic intake.
- Supervised 10+ member committees across major organization projects, ensuring project quality and timely delivery through cross-functional coordination.